

Indian Sign Language Recognition Using Transfer Learning with MobileNetV2

Abstract

This project implements an Indian Sign Language (ISL) recognition system using Transfer Learning with MobileNetV2 in TensorFlow/Keras on Kaggle GPU. The dataset contained 42,745 images across 35 classes (A–Z and 1–9). Images were loaded, split into training, validation and test datasets and evaluated using a confusion matrix and classification metrics.

Objectives

1. Build an ISL recognition system.
2. Apply Transfer Learning.
3. Train using Kaggle GPU.
4. Evaluate using unseen test data.
5. Visualize results using a confusion matrix.

Tools and Technologies

Kaggle Notebook, TensorFlow, Keras, MobileNetV2, Python, Matplotlib, Seaborn and Scikit-learn.

Dataset Details

Dataset: Indian Sign Language Dataset
Total Images: 42,745
Classes: 35
Image Size: 224×224
Batch Size: 32

Methodology

Dataset upload → Dataset path correction → Data loading → Train/Validation/Test split → Transfer Learning model construction → Training → Evaluation → Confusion Matrix.

Transfer Learning

Transfer Learning reuses a pretrained model on a new task. MobileNetV2 pretrained on ImageNet was used. The base model was frozen and a custom classifier head with GlobalAveragePooling, Dropout and Dense layers was added.

Model Architecture

Input Image → MobileNetV2 → GlobalAveragePooling2D → Dropout(0.3) → Dense Softmax Output Layer

Challenges Encountered

Initially only one class was detected because an extra folder level existed in the dataset path. GPU initialization and dataset path corrections were required.

Results

Final Test Accuracy: 100.0%
Final Test Loss: 1.3825654605170712e-05

Conclusion

Transfer learning significantly reduced training time and produced high performance for ISL recognition. Future improvements include fine-tuning and real-time webcam integration.